

Unit 1.3 - Series

1 Definition and Notation

A **series** is the sum of a list of terms (whether finite or infinite). Given the list $a_1, a_2, a_3, \dots, a_n$, a series is expressed as:

$$S = a_1 + a_2 + a_3 + \dots + a_n$$

The series can be written in a more compact form using sigma notation. The **sigma notation** represents the sum of terms in a sequence as follows:

$$\sum_{i=1}^n a_i,$$

where:

- $\sum$ (sigma) denotes the summation symbol
- i is the index of the list, starting from the integer 1 and incrementing by 1 until it reaches n
- a_n represents the general term of the sequence

If the series is infinite, it is expressed as:

$$\sum_{i=1}^{\infty} a_i,$$

2 Examples of Series

- **Sum of the first n natural numbers:** The series is:

$$1 + 2 + 3 + \dots + n.$$

The formula for the sum of the first n natural numbers is:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}.$$

- **Sum of the first n odd numbers:** The series is:

$$1 + 3 + 5 + \cdots + (2n - 1).$$

The formula for the sum of the first n odd numbers is:

$$\sum_{i=1}^n (2i - 1) = n^2.$$

- **Sum of the first n even numbers:** The series is:

$$2 + 4 + 6 + \cdots + 2n.$$

The formula for the sum of the first n even numbers is:

$$\sum_{i=1}^n 2i = n(n + 1).$$

- **Sum of an infinite geometric series:** A geometric series has the form:

$$a + ar + ar^2 + \cdots$$

The formula for the sum of an infinite geometric series is:

$$\sum_{i=0}^{\infty} ar^i = a \frac{1}{1 - r}, \text{ for } r < 1.$$

For example,

$$\sum_{i=0}^{\infty} \frac{1}{2^i} = 2$$

and

$$\sum_{i=1}^{\infty} \frac{9}{10^i} = 0.\bar{9} = 1$$